IMPE LaTeX System

User Manual

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1 Introduction

The IMPE LaTeX System is a modular XeLaTeX framework for reusable layouts, font routing, multilingual typesetting, and optional document features.

The system has three principal author-facing components: *layouts*, *fonts*, and *features*. A layout describes the principal visual organization of a document. Font families provide global or local font behavior, including script-aware routing where necessary. Features provide optional functionality such as mathematics, hyperlinks, citations, tables, or images. These parts can be selected

independently or combined through one unified template interface.

The normal use of IMPE is intentionally short. A wrapper class can establish a sensible document layout and a default global font setup, after which a user loads the additional families or features required by the document. When more explicit control is wanted, the same components can be declared in a single `\UseTemplateSet` call.

XeLaTeX is the primary document engine for IMPE. This choice allows the system to rely on Unicode text, OpenType fonts, `fontspec`, and XeTeX's inter-character mechanisms while retaining the familiar LaTeX document model. Specialized writing systems can additionally use the bundled XSR renderer, which runs from the XeLaTeX workflow. Compile IMPE documents with XeLaTeX.

This manual is a user-oriented introduction. Detailed subsystem documentation is maintained in the repository under `docs/`. In particular, the font, layout, feature, and system documents describe registration fields and implementation details that would be unnecessarily repetitive here. This English manual is the reference edition: translated manuals should follow its structure and technical content.

A visual survey is available in the repository Showcase PDF. The corresponding PDF for this edition is available as the v1.1.0 release asset. The Showcase PDF is also embedded in Appendix B. It is intended to demonstrate the range of scripts, writing directions, layouts, and feature combinations that motivated the system.

2 Installation

2.1 Distribution forms

IMPE 1.1.0 is distributed in three forms. They share the same public class/package names and the core XeLaTeX runtime, but differ in font assets and installation workflow.

`Full` The complete managed release, including the runtime, compatibility entry points, vendored XSR, and only those local font files approved for redistribution and actually present when the archive is built. A public Full archive is not a promise that every catalogued font is bundled.

`Core` The managed runtime, compatibility entry points, vendored XSR and installers, without a local font library. Existing TeX Live and system fonts can be resolved directly; other required files must be supplied.

`CTAN` The `impe-framework.zip` source and runtime distribution rooted at `impe-framework/`. It includes canonical `impe*` entry points, the vendored XSR sources, and documentation resources, but no font binaries. CTAN installation itself does not run `pip` or create a Python environment.

The project is designed to keep the package runtime separate from licensing and availability decisions about large third-party font collections. The final public Full archive uses an explicit reviewed font allowlist.

2.2 Installing Full or Core

Direct installation places the managed package under `TEXMFHOME/tex/latex/impe/`. Its canonical entry points are `impe.sty`, `impeart.cls`, `impebook.cls`, `impereport.cls` and

`impebeamer.cls`; the `_zh` wrappers and historical `next*` names are retained for compatibility in direct releases.

The provided installers work from an extracted Full/Core release or from a complete source checkout. Typical entry points are:

```
# Windows PowerShell, in an extracted release:
.\install.bat
# Windows PowerShell, in a source checkout:
.\scripts\install.ps1

# Unix, in an extracted release:
sh ./install.sh
# Unix, in a source checkout:
sh ./scripts/install.sh
```

The Windows installer accepts `-TexmfRoot` and `-SkipXsrPython`; the Unix installer accepts an optional TEXMF destination as its positional argument. When its installation completes, refresh the TEX filename database if the installer could not run `mktexlsr`. A package installed through the TEX distribution's own package manager follows that distribution's installation rules instead.

2.3 XSR and the Python renderer

IMPE bundles XSR **1.3**. Full and Core installers copy its TEX integration and normally set up a *private* Python virtual environment for its renderer, without installing XSR into the user's global Python environment. Dynamic Egyptian hieroglyphic composition, native Khitan stacking, and XSR vector glyph processing may require that renderer.

The normal managed installation requires Python **3.11 or newer**, `venv`, and `pip`; the initial dependency installation may require network access. A TEX-only installation can skip automatic Python setup:

```
# Windows:
.\scripts\install.ps1 -SkipXsrPython
# Unix:
IMPE_SKIP_XSR_PYTHON=1 sh ./scripts/install.sh
```

The TEX-only option installs the TEX runtime, not a working dynamic Python renderer; configure the renderer separately if using XSR. Dynamic XSR jobs need an allowed XeLaTeX shell-escape invocation or the XSR preprocessing route. Ordinary documents not requesting those renderers do not start Python. The XSR script backends also need the real readable TTF/OTF files used to produce their glyph outlines: an installed font family name by itself is insufficient for the Python font-file reader.

2.4 Font resources and local configuration

For a registered face IMPE first checks the corresponding local/bundled file, then attempts resolution through the TEX font system or operating system. Core and CTAN users do not have to

copy an already resolvable system font into the IMPE repository. Other registered fonts need the necessary files supplied by the user, subject to their respective licenses.

The normal local font root is `assets/fonts/`. To point at a separately managed font library, put the following in `impe.local.tex`, or issue the command from the document preamble:

```
\SetCatalogFontRoot{D:/Fonts/IMPE}
```

The path is only an example. Full installations can provide an automatically managed local root. Unresolved ordinary fonts default to a soft warning and LaTeX fallback; that fallback does not substitute for the actual outline file required by an XSR script renderer.

2.5 A quick installation check

On a system with Libertinus available, the following checks that the wrapper, registry and local font command can be found:

```
\documentclass{impeart}
\UseFont{libertinus}
\begin{document}
IMPE is ready. \LIB{Local Libertinus text.}
\end{document}
```

This checks normal XeLaTeX operation; it does not test XSR/Python or every optional feature. See the project README and the installation checklist for separate XSR and release validation.

3 Quick Start

The simplest IMPE workflow is to choose a wrapper class, select any desired font family or feature, and write the document normally. IMPE provides a reusable setup layer around ordinary LaTeX document structure.

A small article can therefore look like this:

```
\documentclass[11pt]{impeart}

\UseFont{libertinus}
\UseFeatures{math,hyperlinks}

\title{A Sample Document}
\subtitle{Typeset with IMPE}
\author{Author Name}

\begin{document}

\maketitle

\section{Introduction}
\LIB{Hello, world.}
```

```
\end{document}
```

The class already supplies the normal English article layout and base font. The mode-free `\UseFont{libertinus}` call uses its preset's automatic dispatcher: it defines the local `\LIB` command and registers a global Latin font candidate. An explicit user font takes precedence over the implicit CMU class default. Meanwhile, `\UseFeatures` activates optional modules. The remainder is ordinary LaTeX.

For a document whose setup is better expressed as a single declaration, the same idea can be written through `\UseTemplateSet`:

```
\documentclass{impeart}

\UseTemplateSet{
  fonts = {libertinus},
  features = {math,hyperlinks}
}
```

A template set may also choose a layout and additional font families. In practice, wrapper classes already supply a suitable default layout, so most documents state the settings that differ from those defaults.

IMPE can also be loaded as a package underneath a standard LaTeX class:

```
\documentclass{article}
\usepackage{impe}

\UseTemplateSet{
  layout = en_doc,
  fonts = {libertinus},
  features = {hyperlinks}
}
```

This form is useful when the underlying class must remain explicit or when every IMPE component should be selected manually. The wrapper classes and the raw package use the same underlying system.

4 Ready-to-use Templates

The repository includes a small collection of starter projects under `templates/`. These are intended for actual writing. Regression fixtures and internal demonstrations live under `tests/`.

The current starter templates are:

`article_en/` English article starter using `impeart`.

`article_zh/` Chinese article starter using `impeart` with `lang=zh`.

`book_en/` English book starter using `impebook`.

book_zh/ Chinese book starter using impebook with lang=zh.

beamer_en/ English presentation starter using impebeamer.

beamer_zh/ Chinese presentation starter using impebeamer with lang=zh.

Each template follows the installed-package style and contains realistic metadata, sectioning, and body structure, together with representative table, figure, or slide material where appropriate. A new project can therefore begin by copying the nearest template directory and editing the document contents using the supplied preamble as a starting point.

The templates deliberately use the canonical impe* entry points and show normal user-facing practice. Development fixtures live under tests/.

5 Document Classes and Public Entry Points

5.1 Canonical wrapper classes and language selection

Four wrapper classes are the standard entry points for new documents: impeart, impereport, impebook and impebeamer. Each class accepts lang=en (the default) or lang=zh. Use the same class name for both languages:

```
% English article
\documentclass[12pt,twoside]{impeart}

% Chinese article
\documentclass[lang=zh,12pt]{impeart}

% Chinese book
\documentclass[lang=zh]{impebook}

% Chinese presentation
\documentclass[lang=zh]{impebeamer}
```

The underlying classes are article, report, book and beamer, respectively. English layouts are en_doc, en_report, en_book and beamer; lang=zh selects zh_doc, zh_report, zh_book and beamer_zh. The English default font is CMU; Chinese documents add Shanggu to the implicit font defaults. An explicit author font selection can override the class defaults.

Other class options, such as 12pt, twoside and the native Beamer aspectratio option, are passed through to the underlying class. The historical impe*_zh and next* class names remain supported as compatibility entry points; they are listed separately in Section 9.

5.2 Using impe.sty directly

The wrapper classes share one implementation and provide convenient defaults. A user may also keep a standard class and load

```
\usepackage{impe}
```

followed by an explicit layout, font, and feature configuration. This is the recommended route when the document class is supplied by a journal, publisher, or another package and must remain the document's class.

5.3 Titles and subtitles

IMPE supports `\subtitle{...}` alongside the standard `\title` and `\author` commands. Wrapper classes apply their own title-block defaults, while the ordinary LaTeX title mechanism remains recognizable. A typical preamble is simply

```
\title{Main Title}
\subtitle{A More Specific Description}
\author{Author Name}
```

followed by `\maketitle` in the document body.

6 The Template Model

6.1 Four setup keys

The principal unified command is `\UseTemplateSet`. Its public setup model is intentionally small:

```
\UseTemplateSet{
  layout = <preset>,
  fonts = {<families>},
  globalfonts = {<families>},
  features = {<features>}
}
```

The four keys have different roles. Global declarations made explicitly by the author have precedence over wrapper-provided defaults; when several user fonts compete for the same primary slot, their declaration order can matter.

`layout` selects a public layout preset appropriate to the active document class.

`fonts` loads families through their registered automatic behavior. This is the normal choice: a family may define a local command, apply a document-wide font, or install range-specific routing.

`globalfonts` explicitly forces the listed families through their global mode or global routing path.

`features` loads optional functional modules.

`mainfonts` is accepted as an alias of `globalfonts`; both keys provide explicit global overrides; `fonts` provides the automatic interface.

The template interface provides one predictable place to express a complete configuration. Wrapper classes already apply their own layout and global-font defaults, so a document may specify an additional feature list alone.

6.2 Single-purpose shortcuts

For focused configuration, IMPE provides public shortcuts corresponding to the individual subsystems:

```
\UseLayout{en_doc}
\UseLayouts{...}

\UseFont{libertinus}
\UseFonts{arabic,tibetan}

\UseFont{libertinus}[global]
\UseLocalFonts{arabic,tibetan} % explicit local override
\UseGlobalFonts{libertinus}    % explicit global override

\UseFeature{hyperlinks}
\UseFeatures{math,hyperlinks,tables}
```

The mode-free font forms are the recommended shortcuts for ordinary use. The local and global aliases remain useful when a document intentionally overrides a family's registered behavior. The unified and single-purpose interfaces are two views of the same public model.

6.3 Compositional templates

An IMPE template combines reusable components. A layout can be reused with different font choices, the same feature set can be used in an article or book, and a script-specific font family can be added while preserving unrelated document functionality. This composition is the central design principle behind the public interface.

For example, the following declarations compose two different documents from the same template interface:

```
% English article: the class provides en_doc and CMU
\documentclass{impeart}
\UseTemplateSet{
  fonts = {libertinus},
  features = {math,hyperlinks}
}

% Chinese book: the class provides zh_book, CMU and Shanggu
\documentclass[lang=zh]{impebook}
\UseTemplateSet{
  fonts = {sanskrit,tibetan},
  features = {citations,index,hyperlinks}
}
```

These are alternative preambles for separate documents. The wrappers already select their language-specific layout and implicit fonts; each author declaration adds only the required fonts and features. Use `layout=` when selecting a different registered layout explicitly.

7 Fonts, Script Routing, and Multilingual Typesetting

For font selection, IMPE offers registered presets and explicit Writing Profiles. Both are supported public APIs in 1.1.0: presets keep their familiar local shortcuts, while profiles pair Unicode coverage with independently selected font resources.

7.1 Simple registered font presets

The easiest way to load an existing registered preset is `\UseFont{id}`; the `fonts` key of `\UseTemplateSet` accepts a list of the same IDs. Omitting the mode follows that preset's *automatic dispatch*, which can enable a local command, a document-wide font binding, and/or Unicode-range routing. The resulting behavior depends on the preset's complete registration, including its global capability and assigned Unicode ranges.

```
\UseFont{libertinus}       % automatic: local command plus global candidate
\UseFonts{sanskrit,tibetan,arabic}
\UseFont{libertinus}[local]   % local declaration only
\UseFont{libertinus}[global]  % explicit global declaration
```

The first call makes `\LIB{...}` available and also registers Libertinus as an explicit global Latin candidate. In a wrapper document it can replace the implicit CMU primary according to the precedence rules below. An explicit `[local]` confines the request to the local declaration. An explicit `[global]` requires a global-capable preset. The singular and plural `\UseLocalFont` and `\UseGlobalFont` forms provide explicit overrides; `\UseMainFont` is an alias for global selection. Their plural forms accept comma-separated ID lists.

A local command, such as `\SA{...}`, applies to its text argument, preserving the script's shaping and special renderer. The generic `\Font{preset}{text}` offers the same forced-local selection without needing an earlier `\UseFont` call. Family commands and the preset interface remain especially convenient for established editorial styles.

7.2 Writing Profiles and Font Resources

The writing model maintains four registries: *Unicode block data*, *Writing Profiles* (codepoint coverage and script behavior), *font resources/packs* (real font faces and styles), and *registered presets* (the familiar, complete font recipes). For example, the `sanskrit` preset specifies Sanskrit typography, while the `devanagari` profile identifies Unicode coverage and script behavior that can be paired with a chosen font resource.

Version 1.1.0 provides 69 curated profiles, 85 resource aliases and all 59 original registered preset IDs. The profile determines Unicode coverage and the appropriate script/renderer policy; a resource override changes the font choice, not the profile's identity. Thus `chinese`, `japanese` and `korean` profiles can share Han codepoints: Unicode alone cannot guess which regional Han form is intended.

A global writing declaration uses:

```
\UseWriting{[shanggu]chinese}
\UseWriting{[tiro]devanagari}
\UseWriting{[naskh]arabic}
```

The general syntax is `\UseWriting{[pack,options]profile}`. Multiple entries may be comma-separated outside brackets and braces. The optional first bracket item selects a registered resource alias; alternatively, `font=` names an installed family and `file=` identifies a font file. Select one of these resource forms per entry. Other keys include `stylemap=native|full`, `writing=horizontal|vertical`, `offset=` and face overrides such as `italic=zhuque`. Global writing is horizontal; the vertical option is available in local rendering where the selected profile supports it.

7.3 Named and anonymous local writing

Use `\DeclareWriting{name}{configuration}` to save a reusable configuration without activating it globally. `\Font{name}[overrides]{text}` forces that configuration locally; `\Writing{configuration}{text}` is its anonymous counterpart. For example:

```
\DeclareWriting{han-note}
  {[shanggu,stylemap=full,italic=zhuque]chinese}
\begin{document}
\Font{han-note}{ }
\Font{sanskrit}{   }
\Writing{[tiro]devanagari}{   }
\end{document}
```

Local selection applies one font resource and renderer to the *entire supplied argument*; it does not automatically subdivide that argument according to the profile's Unicode ranges. Nested explicit local calls can override the outer selection. A local configuration may also specify only `font=` or `file=` when no script-specific Profile behavior is needed.

The `stylemap=native` policy favors the resource's genuine style faces; `stylemap=full` additionally permits the documented cross-family stylistic mappings. Several multi-face Western packs default to the full mapping. Style selection and missing-character recovery are separate operations: an available Bold face may still lack a particular character. The complete option syntax and resource details appear in `docs/USEWRITING-1.0.5.md`; the appendix below includes the registered names.

7.4 Global precedence and automatic fallback

A wrapper's implicit CMU (and, for `lang=zh`, Shanggu) setup supplies the default font pool, but *explicit user declarations take priority* over those class defaults. Within one priority tier, the first declared Latin global primary normally remains primary. A later global registration does not automatically replace the earlier explicit primary. For CJK cross-profile conflicts within a tier, the existing Chinese–Japanese–Korean priority is retained. Explicitly selecting a previously implicit default can promote it to the user tier.

For example, after `\documentclass{impeart}`, calling `\UseFont{libertinus}` explicitly lets Libertinus supersede the class-provided CMU primary. But in

```
\UseGlobalFonts{cmu,libertinus}
```

the first explicit Latin primary, CMU, remains primary. This difference is intentional. It also explains why an *Auto* table entry describes registration capabilities and effects, not an unconditional promise that every requested font becomes the document's main face.

Automatic missing-glyph fallback currently applies to CJK only. The xeCJK backend searches the active resource's declared faces using the Serif/Sans × Regular/Bold × Upright/Italic (2×2×2) style matrix. It then considers available registered resources of the same Writing Profile, followed by other CJK profiles. The global fallback resource pool places explicit user resources before implicit class defaults and maintains first-registration order within a tier. Candidates with a valid face but a missing glyph can be skipped; late registrations refresh the relevant fallback chains.

When a font declaration cannot be resolved, IMPE's soft fallback supplies the configured substitute. Glyph-by-glyph automatic recovery is scoped to CJK, preserving whole-run OpenType shaping for other scripts.

7.5 Unicode-range and CJK routing

A script-specific global binding can claim selected Unicode blocks while unrelated text stays with its existing font. Existing local commands take precedence within their argument and suppress the outer automatic range router there. Leaving their scope restores global routing.

For CJK, IMPE uses xeCJK main/sans/mono channels. Shared Han characters retain the active document CJK selection, while kana and Hangul can be claimed as distinct script ranges. Use local selection such as `\JP{...}` or `\KR{...}` when a regional form of a shared Han ideograph must be chosen explicitly. Overlapping global Writing Profiles follow the existing route-ownership rules.

7.6 Shaping, line breaking and writing directions

OpenType shaping for Indic, Arabic and other complex scripts remains the responsibility of XeTeX and the selected OpenType fonts. IMPE supplies script, language and feature settings and selected script-specific breaking behavior: Sanskrit Devanagari akshara boundaries, Tibetan tsheg boundaries and Thai dictionary-based line breaks are supported in their documented routes. Right-to-left behavior is attached to appropriate presets and renderers. For Mongolian, Manchu and Old Uyghur, IMPE supplies local rotated vertical forms; these are distinct from page-wide writing-direction changes.

7.7 XSR 1.3: Egyptian, Khitan and inline glyph images

IMPE bundles XSR 1.3 for Egyptian composition, Khitan Small Script stacking and inline image glyphs. The Egyptian and Khitan font presets activate their respective script backends; `glyphs` provides independent inline image registration.

Egyptian hieroglyphs Load `\UseFont{egyptian}`, then use the renderer's supported Unicode hieroglyphic controls or `\IMPEEgyptianText{...}` where explicit selection is preferable.

Khitan Small Script Load `\UseFont{khitan_small}` and use `\KHS{...}`, the default *native XSR stack* renderer. `\KHSstack{...}` and `\KHSstackblock{...}` remain explicit aliases. Unicode cluster separators and U+16FE4 fillers are retained; XSR 1.3 uses normal host-text interword spacing and vertically centers stacks of differing heights.

Registered inline images Load `\UseFeature{glyphs}` and register glyph assets with `\IMPEGlyphRegister`; insert them through `\IMPEGlyph`. Its image rendering is independent of Egyptian and Khitan script detection.

Dynamic rendering requires Python 3.11+, the installed private XSR runtime or an equivalent configured renderer, XeLaTeX shell escape (or XSR preprocessing), and readable source/font files.

XSR 1.1 and later can preserve the original linear Unicode source of Egyptian quadrats and Khitan clusters via PDF `/ActualText`, including controls and separators, while drawing the stacked visual output. This preserves Unicode source text alongside visual stacking; actual copy behavior depends on the PDF reader. The mechanism applies to these XSR text compositions, while ordinary equations and image glyphs retain their respective PDF text behavior.

7.8 Further font documentation

The complete family catalog, source/font resolution, writing options, style-face mapping, compatibility presets, XSR requirements and script-specific notes are maintained in `docs/FONTS.md` and `docs/USEWRITING-1.0.5.md`. The public Showcase illustrates the actual rendered output.

8 Layouts and Features

8.1 Publication layout presets

Layouts coordinate page dimensions, text rhythm, headers and (where applicable) chapter-opening or Beamer behavior. They provide publication defaults that can be adjusted through native LaTeX packages. The canonical presets are:

`en_doc` / `zh_doc` English/Chinese article: A4, 29 mm left/right, 28 mm top and 32 mm bottom margins.

`en_report` / `zh_report` English/Chinese report: A4, 31 mm left, 29 mm right, 29 mm top and 34 mm bottom margins, with independent report running heads.

`en_book` / `zh_book` English/Chinese book: ISO B5 (176×250 mm), two-sided with 19 mm inner, 31 mm outer and 3 mm binding offset.

`beamer` / `beamer_zh` Madrid theme, 4:3 canvas, no navigation symbols; explicit native `aspectratio` class options remain effective.

Article and report have independent A4 geometry and typography. IMPE checks each preset against the underlying document class. The historical `report` layout ID is recorded in the compatibility section.

The optional `beamer_clean` and `beamer_clean_zh` presets select the minimalist presentation appearance introduced in 1.0.6. The default `beamer` presets use Madrid 4:3; set `aspectratio=169` in the class options for widescreen slides.

8.2 Native layout overrides and local settings

The canonical article, report and book classes load `geometry` and `fancyhdr`. For detailed page geometry and running heads, use their normal LaTeX interfaces after the class and any explicit `\UseLayout` declaration:

```
\documentclass{impereport}
\geometry{a4paper,left=30mm,right=30mm}
\fancyhead[R]{A short running title}
```

IMPE also provides `\LayoutSetup` and `\LayoutSet` for registered component settings. Use `LayoutScope` for paragraph-local text adjustments, within the document body:

```
\documentclass{impeart}
\LayoutSetup{page.inner=25mm,page.top=24mm}
\begin{document}
\begin{LayoutScope}{text.par-indent=2em}
A paragraph with a scoped indentation setting.
\end{LayoutScope}
\end{document}
```

The header presets support `running`, `title`, `classic`, `minimal` and `report`. Native Beamer configuration remains available for slides. For historical 1.0.5 layout parameters, see the compatibility section, Section 9, and `docs/LAYOUTS.md`.

8.3 Feature modules

Features load curated packages and retain their familiar native interfaces. The public feature IDs in IMPE 1.1.0 are:

`math` AMS-oriented mathematics, theorem environments, optional math-font selection.

`hyperlinks` Unicode-aware links and PDF bookmarks, stable destinations and footnote links.

`citations` `biblatex`/`biber` citations and citation-style presets.

`index` Native-first `imakeidx`/`xindy` indexing and exact occurrence links in IMPE's default index.

`tables` Standard tables with `booktabs`, `longtable`, `tabularx` and related packages.

`image` `graphicx`, captions, subfigures.

`drawing` TikZ/PGF, pgfplots (compatibility 1.18) and forest for diagrams, plots and trees.

`glyphs` Standalone inline image glyph registration via XSR's vector backend, separate from script font detectors.

`headers` Configurable running heads using `fancyhdr`.

The retained `lists_envs` feature provides the `ExampleBlock` environment for existing documents; the aliases `bib` and `header` are also supported. See Section 9 for these compatibility APIs.

```
\UseFeatures{math,hyperlinks,tables,image,drawing,glyphs}
```

Feature modules can be combined; repeated activation is ignored.

8.4 Mathematics and citations

The `math` feature defines `theorem`, `lemma`, `proposition`, `corollary`, `definition`, `example` and `remark`. Computer Modern mathematics remains the default when only a text family is selected. To choose another math alphabet before loading the feature:

```
\UseMathFont{libertinus}
\UseFeature{math}
```

This selects Libertinus Math through `unicode-math`. The `citations` feature uses native `biblatex` commands including `\addbibresource`, `\textcite`, `\parencite` and `\printbibliography`. The `\UseCitationStyle` presets include APA, GB/T 7714, numeric and author–year. Citation options are configured before the feature loads.

8.5 Indexes: native `imakeidx` first

New documents use native `\index`, `\makeindex` and `\printindex`, with IMPE adding stable destinations for ordinary entries in its default index. When IMPE creates its default index, each ordinary native `\index` mark receives a unique PDF destination for that occurrence, even when page numbering has been reset or several marks fall on the same physical page.

```
\UseFeature{index}
\begin{document}
Buddhism\index{Buddhism} and
Mahayana\index{Buddhism!Mahayana} are discussed here.
Later Buddhism\index{Buddhism} can be marked again.
\printindex
\end{document}
```

If no native default index already exists, the feature initializes one using xindy. Run the indexer in the document output directory and then XeLaTeX twice:

```
texindy -L english -C utf8 -M <job>-impe <job>.idx
```

IMPE respects a pre-existing user default index. Named indexes use standard declarations such as `\makeindex[name=people,title={People},program=xindy]` with `\index[people]{...}` and `\printindex[people]`. The exact occurrence-link guarantee is limited to plain entries in IMPE's initialized default index. Native formatting encapsulators, `see`/`seealso`, range delimiters and named indexes retain their own indexing and hyperlink semantics. `\Term` and its first-use behavior remain available for compatible first-use terminology workflows.

8.6 Tables, figures, drawings and inline glyphs

Write new tables using `table`, `tabular`, `longtable`, `booktabs` rules and `tabularx` columns; use `figure`, `\includegraphics` and `subfigure` for images. The former `TableBook*`, `OneImage*` and `PanelFigure*` convenience layers remain supported for existing documents. For native vector diagrams, use `tikzpicture`, `axis` or `forest` after loading `drawing`.

The independent image glyph interface is:

```
\UseFeature{glyphs}
\IMPEGlyphRegister[logical={unencoded sign},description={rubbing}]
  {bs-042}{images/bs-042.svg}
\begin{document}
Before \IMPEGlyph{bs-042} after.
\IMPEGlyph[scale=1.2,raise=1pt,trim={1pt 0pt 1pt 0pt}]{bs-042}
\end{document}
```

The image path must exist and be available to the renderer. The optional `logical` and `description` metadata do not insert invisible duplicate PDF text; `trim` clips the placed box, not the source asset.

8.7 Headers and hyperlinks

The `headers` feature can derive a running title from the first line of `\title`, with `\HeaderTitle{...}` for a shorter fixed form. The selected style determines the use of section/chapter running information. Complex overrides should use native `fancyhdr` commands.

The `hyperlinks` feature loads `hyperref` and `bookmark` with Unicode-aware PDF settings, stable destinations and linked contents. Native `\label`, `\ref`, `\href` and `\url` remain the standard interfaces. See `docs/FEATURES.md` for technical details.

9 Compatibility and Migration

IMPE 1.x establishes `impe*` as the canonical public naming scheme. New documents use `impe.sty`, `impeart`, `impebook`, `impereport` or `impebeamer`, with `lang=zh` where desired. Historical `next*` names and `_zh` wrappers remain usable. Their exact mappings are to the corresponding canonical class with the `lang=zh` option: for example, `\documentclass{impeart_zh}` becomes `\documentclass[lang=zh]{impeart}`. The same mapping applies to `impereport_zh`, `impebook_zh` and `impebeamer_zh`. The `nextart`, `nextreport`, `nextbook`, `nextbeamer`, `nextsystem` and their available `_zh` variants are legacy package/class names in Full/Core installations.

The existing `\UseFont` IDs, local shorthand commands (for example `\SA`, `\JP`, `\MOv`), and explicit `[local|global]` modes are supported. The Writing Profile/Resource interface adds options rather than altering the old preset namespace or rewriting its compatibility recipes.

The recommended indexing interface is now native-first. The `\Term[options]{display}[description]` and `\IndexTitle` helpers retain their formatting and first-occurrence semantics. Native `\index` continues to register every ordinary occurrence. For new first-use terminology or glossary definitions, consider the separate `glossaries-extra` package directly.

The compatible table helpers `TableInlineFit`, `TableLong`, `TableBook`, `TableBookX`, `TableBookNotes`, `NiceBooktable*`, `OneImage`, `OneImageInline`, `PanelFigure`, `PanelFigure*`, and `\Panel` helpers remain available. Prefer native LaTeX tables, figures and subfigures in new sources. The `lists_envs` `ExampleBlock` environment is also retained for compatibility.

For a document that needs the 1.0.5 layout parameters, select `\UseLegacyLayout` immediately after the canonical document class, before author overrides:

```
\documentclass[lang=zh]{impebook}
```

`\UseLegacyLayout`

This compatibility helper chooses the historical layout for the active class and language. The earlier `report` layout ID and `legacy_*` layout IDs remain available; font versions and other dependencies can still affect pagination.

The direct installer recognizes managed installations from the 0.1.x series under `tex/latex/nextsystem/`. It installs the canonical runtime under `tex/latex/impe/`, migrates recognized local overrides when appropriate, removes only files known to belong to the old managed runtime, and preserves other user-created files. New persistent configuration should use `impe.local.tex`.

Compatibility preserves public entry points and documented commands. Font and layout updates may change line breaks or pagination.

10 Package Structure and Extension Points

The repository separates public entry points, stable mechanisms, registrations, specialized implementations, and local resources:

```
package/   public package and class entry points
core/      stable subsystem mechanisms
catalog/   public ids and registration data
modules/   extendable or specialized implementations
assets/    local runtime resources such as fonts
scripts/   installation and release tooling
manual/    manual sources
docs/      detailed subsystem documentation
```

10.1 The stable core

The `core/` layer contains stable shared mechanisms for new font families, layout presets, and features. It implements the font registry and routing machinery, the layout registry and compatibility checks, the feature loader, and related shared behavior.

This structure allows new ordinary OpenType fonts to be registered through catalog data and new features to be connected through the feature catalog, while the shared routing and loading mechanisms stay centralized in the core.

10.2 Catalogs

The `catalog/` layer assigns public ids and metadata. It contains the central registrations for fonts, layouts, and features. For many extensions, this is the correct place to begin.

An ordinary font family using standard `fontspec` shaping can usually be expressed entirely as a catalog registration. Likewise, a public layout preset is assembled from existing internal components, and a feature id maps to its implementation module. Keeping those declarations centralized makes the public surface auditable in one place.

10.3 Modules

The `modules/` layer is reserved for functionality that genuinely needs its own implementation: feature modules, specialized script support, or reusable internal layout components. Specialized

modules implement behaviors that need dedicated code; font names and face declarations belong in catalog registrations.

This rule is particularly important for multilingual support. Ordinary OpenType shaping and catalog metadata cover the majority of scripts; specialized modules serve writing systems and rendering strategies with additional requirements.

10.4 Extending IMPE

Developers extending the system should begin with the detailed registration interfaces in `docs/SYSTEM.md` and then consult the relevant subsystem document: `docs/FONTS.md`, `docs/LAYOUTS.md`, or `docs/FEATURES.md`.

Choose a catalog registration for ordinary additions and a specialized module for behavior that needs its own implementation.

11 Further Documentation and Feedback

This manual is the normal starting point for using IMPE. The repository contains more detailed technical documentation for users who need a complete reference or who want to extend the system.

`docs/SYSTEM.md` architecture, public entry layers, template setup, release model, and extension boundaries.

`docs/FONTS.md` family registration, local and global modes, routing, fallback, shaping, writing behavior, vertical support, and XSR integration.

`docs/LAYOUTS.md` public layout presets, compatibility targets, and the component model from which presets are built.

`docs/FEATURES.md` public feature ids and the commands, environments, package stacks, and compilation requirements supplied by each module.

`CHANGELOG.md` changes in published versions.

Showcase the full visual showcase is available from the project repository.

The project repository is available at

`github.com/KelvinYangBK67/IMPE-LaTeX-System`

Bug reports, feature requests, documentation corrections, portability reports, and other concrete feedback are welcome through GitHub Issues:

`github.com/KelvinYangBK67/IMPE-LaTeX-System/issues`

When reporting a typesetting problem, it is useful to include a minimal source file, the TEX engine and TEX Live version, the IMPE version, and the relevant font information. Reports concerning a local or privately supplied font should also distinguish between a font-coverage problem and an IMPE routing problem where possible.

The English manual is the authoritative manual text. Other language editions may adapt prose and terminology to their own technical conventions, but their commands, examples, and description of IMPE behavior should remain synchronized with this edition.

A Quick Reference

This appendix summarizes the *current* 1.1.0 public authoring interface. It distinguishes convenient registered presets, explicit Writing Profiles and font resources, native feature APIs, and compatibility-only helpers. Detailed parameter lists and extension contracts live under `docs/`.

A.1 Canonical classes and language options

The same four wrapper classes serve English (`lang=en`, the default) and Chinese (`lang=zh`) documents. Use `\documentclass[lang=zh]{impeart}`, for example, to select the Chinese article layout and implicit font defaults.

Class	Default English	With `lang=zh`
`impeart`	`en_doc`; A4; CMU	`zh_doc`; A4; CMU and Shanggu
`impereport`	`en_report`; A4; CMU	`zh_report`; A4; CMU and Shanggu
`impebook`	`en_book`; ISO B5; CMU	`zh_book`; ISO B5; CMU and Shanggu
`impebeamer`	`beamer`; Madrid 4:3; CMU	`beamer_zh`; Madrid 4:3; CMU and Shanggu

For an externally supplied document class, load `\usepackage{impe}` and declare the desired layout, fonts and features. The historical `impe*_zh` and `next*` entry points are listed with the compatibility interfaces below.

A.2 Core setup and font presets

Public command	Role
`\UseTemplateSet{...}`	Declarative layout/fonts/globalfonts/features setup.
`\UseFont{id}[mode]`	Registered preset: Auto when omitted, or explicit `local`/`global`.
`\UseFonts{ids}[mode]`	The same operation for a comma-separated preset list.
`\UseGlobalFont`, `\UseGlobalFonts`	Explicit global preset binding; `\UseMainFont` and `\UseMainFonts` are aliases.
`\UseLocalFont`, `\UseLocalFonts`	Explicit local-only preset loading.
`\UseLayout{id}`	Select a class-compatible layout preset.

Public command	Role
\UseLayouts{ids}	Apply a list of layout presets.
\UseFeature{id}	Load a feature module.
\UseFeatures{ids}	Load several feature modules.
\SetCatalogFontRoot{path}	Override the local font-file root.
\subtitle{text}	Set an optional document subtitle.

For an overview of all four \UseTemplateSet keys, see the Template Model section. The keys are layout (one preset), fonts (automatic preset dispatch), globalfonts (explicit global binding) and features (feature IDs). mainfonts is an alias of globalfonts. User-explicit global choices outrank wrapper defaults; within a tier, first-registered Latin primaries prevail. CJK has additional regional arbitration.

A.3 Writing Profiles and forced-local text

The following examples use the new 1.0.5-and-later writing model:

```
\UseWriting{[shanggu]chinese}
\DeclareWriting{han-note}
  {[shanggu,stylemap=full,italic=zhuque]chinese}
\Font{han-note}{ }
\Font{sanskrit}{   }
\Writing{[tiro]devanagari}{   }
```

Public command	Role
\UseWriting{config}	Activate global profile/range and font-resource binding.
\DeclareWriting{name}{config}	Store a reusable configuration without activation.
\Font{name}[options]{text}	Force one named preset/configuration locally.
\Writing{config}{text}	Anonymous forced-local configuration.

A configuration normally takes [pack,options]profile. Use font= for resolvable family names, file= for literal paths, and a registered pack for known resources. stylemap=native|full, named face overrides, and writing=horizontal|vertical are among the available settings. Global vertical writing is not supported. The complete Profile and Resource/Pack ID catalogs appear in Sections A.7 and A.8; see docs/USEWRITING-1.0.5.md for option details.

A.4 Layouts and header controls

Preset or command	Role
en_doc, zh_doc	Article A4 typography and page geometry.
en_report, zh_report	Independent A4 report presets.
en_book, zh_book	ISO B5 book presets.
beamer, beamer_zh	Madrid 4:3 presentation presets.
beamer_clean, beamer_clean_zh	Optional minimalist presentation look.
\LayoutSetup{keys}	Adjust supported layout component keys.
\LayoutSet{key}{value}	Set one supported layout key.
LayoutScope	Environment for paragraph-local layout settings.
\HeaderStyle{value}	running, title, classic, minimal or report.
\HeaderTitle{text}	Override the fixed running title.

Native \geometry, fancyhdr, size commands and Beamer customization remain preferable for detailed overrides.

A.5 Public feature identifiers and preferred APIs

ID	Preferred functionality
math	AMS packages; theorem environments; optional \UseMathFont.
hyperlinks	Native hyperref/bookmark links and references.
citations	Native biblatex citations and bibliography.
index	Native \index, \makeindex, \printindex; exact ordinary occurrences in IMPE's default index.
tables	Native tables, booktabs, longtable, tabularx.
image	Native figures, \includegraphics, subfigures.
drawing	tikzpicture, axis, forest.
glyphs	\IMPEGlyphRegister, \IMPEGlyph for image glyphs.
headers	Running headings and native fancyhdr.

Useful settings include \UseCitationStyle, \SetCitationBiblatexOptions, \UseMathFont, and \HeaderStyle. For IMPE's initialized default xindy index, run texindy -L english -C utf8 -M <job>-impe <job>.idx between XeLaTeX passes. The egyptian and khitan_small font presets activate their XSR script renderers.

A.6 Registered font preset identifiers and local commands (59)

The 59 established \UseFont IDs remain supported, and the mode-free preset API is recommended for straightforward font selection. The **Auto** column records the automatic loader's

effects, which can include local commands, global candidates and Unicode routing beyond the catalog's `defaultmode` metadata:

`G` Global font registration only.

`L` Local selection/command without automatic global routing.

`L+G` Local access plus a global font candidate.

`L+R` Local access plus eligible Unicode-range routing.

Automatic route ownership can depend on previously declared fonts, global precedence and range conflicts. For example, `\UseFont{libertinus}` registers both local and global capabilities even though its catalog `defaultmode` is `local`. An explicitly forced `[local]` or `[global]` request has different semantics. A registered global candidate is not guaranteed to become the effective primary font.

Family ID	**Auto**	**Local command**	**Typical role**
`cmu`	G	—	Computer Modern Unicode base family.
`noto`	L+G	`\NOT`	General Noto Latin family.
`times`	L+G	`\TIM`	Times/Arial/Consolas system-family route.
`gentium`	L+G	`\GEN`	Gentium Plus.
`charis`	L+G	`\CHA`	Charis SIL.
`libertinus`	L+G	`\LIB`	Libertinus serif/sans/mono family.
`mlmodern`	L	`\MLM`	Latin Modern / MLModern local route.
`anatolian`	L	`\CA`	Carian / Anatolian script material.
`coptic`	L	`\CO`	Coptic.
`bopomofo`	L	`\ZY`	Bopomofo.
`cuneiform`	L	`\CU`	Cuneiform.
`glagolitic`	L	`\GL`	Glagolitic.
`italic`	L	`\OI`	Old Italic.
`hungarian`	L	`\OH`	Old Hungarian.
`runic`	L	`\RU`	Runic.
`armenian`	L	`\HY`	Armenian.
`georgian`	L	`\KA`	Georgian.
`hindi`	L+R	`\HI`	Hindi Devanagari with registered range routing.

Family ID	Auto	Local command	Typical role
sanskrit	L+R	\SA	Sanskrit Devanagari with Sanskrit-aware routing.
devanagari	L	\DEV	Generic Devanagari shaping.
tamil	L	\TA	Tamil.
brahmi	L	\BR	Brahmi.
tibetan	L+R	\TI	Tibetan with tsheg-aware line-breaking behavior.
segoe	L	\SEG	Segoe Historic local route.
thai	L	\TH	Thai.
arabic	L	\AR	Arabic; serif/sans and alternate stylistic faces.
urdu	L	\UR	Urdu Nastaliq.
aramaic	L	\IA	Imperial Aramaic.
nabataean	L	\NB	Nabataean.
hebrew	L	\HE	Hebrew.
syriac	L	\SY	Syriac.
syriac_eastern	L+R	\SYE	Eastern Syriac with registered range routing.
kharosthi	L	\KH	Kharosthi.
egyptian	L	\EG	Egyptian hieroglyphic composition (XSR; also \IMPEEgyptianText).
khitan_small	L	\KHS	Khitan Small Script with its specialized composer.
pahlavi_parthian	L+R	\PAR	Inscriptional Parthian.
pahlavi_inscriptional	L+R	\PAH	Inscriptional Pahlavi.
pahlavi_psalter	L	\PSP	Psalter Pahlavi with specialized support.
avestan	L	\AV	Avestan.
manichaean	L	\MA	Manichaean.
phoenician	L	\PH	Phoenician.
samaritan	L	\SM	Samaritan.
sogdian	L	\SG	Sogdian.
sogdian_old	L+R	\SGO	Old Sogdian with registered range routing.
chinese_simplified	L+R	\SC	Simplified Chinese CJK family.

Family ID	Auto	Local command	Typical role
chinese_traditional	L+R	\TC	Traditional Chinese CJK family.
japanese	L+R	\JP	Japanese; automatic routing preserves shared Han priority.
wenjin	L+R	\WJ	WenJin Mincho with CJK fallback chain.
shanggu	G	—	Traditional-Chinese-oriented global CJK family.
sim	G	—	Simplified-Chinese-oriented system CJK family.
korean	L+R	\KR	Korean/Hangul family.
tangut	L+R	\TG	Tangut with CJK-aware automatic routing.
mongolian	L	\MO	Mongolian with built-in vertical layout route.
mongolian_baiti	L	\MOb	Mongolian Baiti with vertical layout route.
manchu	L	\MC	Manchu with vertical layout route.
turkic	L	\OT	Old Turkic.
uyghur	L	\UY	Old Uyghur with vertical/rotated layout route.
vietnamese_quocngu	L	\VI	Vietnamese Quốc Ngữ.
vietnamese_hannom	L	\HN	Vietnamese Hán-Nôm / CJK text.

A.7 Curated Writing Profile identifiers (69)

Use a Profile ID with \UseWriting for global range selection or with \Writing and named declarations for local rendering. These identifiers follow the maintained Unicode codepoint composition in core/fonts/impe-fonts-usewriting.tex. The coverage column lists the registered scope keys; several profiles combine multiple blocks or selected portions of a block.

```
\UseWriting{[shanggu]chinese}
\UseWriting{[tiro]devanagari}
\Writing{[naskh]arabic}{    }
```

Default resource is the resource alias selected when a Profile supplies its own font default. A dash means the Profile defines Unicode coverage and script behavior but leaves the font resource to an explicit [pack], font= or file= choice. These profiles can be used with a suitable user-provided font.

Profile ID	Default resource	Coverage key(s)
basic	basic	basic
latin	basic	latin
greek	basic	greek-only
cyrillic	basic	cyrillic
armenian	armenian	armenian
georgian	georgian	georgian
hebrew	hebrew	hebrew
arabic	naskh	arabic
syriac	syriac	syriac
samaritan	samaritan	samaritan
aramaic	aramaic	imperial-aramaic
nabataean	nabataean	nabataean
phoenician	phoenician	phoenician
avestan	avestan	avestan
pahlavi-parthian	pahlavi-parthian	inscriptional-parthian
pahlavi-inscriptional	pahlavi-inscriptional	inscriptional-pahlavi
pahlavi-psalter	pahlavi-psalter	psalter-pahlavi
manichaean	manichaean	manichaean
sogdian	sogdian	sogdian
old-sogdian	old-sogdian	old-sogdian
old-uyghur	old-uyghur	old-uyghur
devanagari	devanagari	devanagari
tamil	tamil	tamil
thai	thai	thai
brahmi	brahmi	brahmi
kharoshthi	kharoshthi	kharoshthi
tibetan	tibetan	tibetan
coptic	coptic	coptic-main, coptic-legacy
glagolitic	glagolitic	glagolitic
old-italic	old-italic	old-italic
old-hungarian	old-hungarian	old-hungarian
runic	runic	runic
old-turkic	old-turkic	old-turkic
carian	carian	carian
cuneiform	cuneiform	cuneiform
egyptian	egyptian	egyptian
khitan-small	khitan-small	khitan-small
tangut	tangut	tangut
han	shanggu	han
chinese	shanggu	chinese
japanese	japanese	japanese

Profile ID	Default resource	Coverage key(s)
korean	korean	korean
kana	japanese	kana
hangul	korean	hangul
bopomofo	iming	bopomofo
mongolian	mnglwhite	mongolian
jurchen	—	jurchen
seal	—	seal
proto-cuneiform	—	proto-cuneiform
bengali	—	Bengali, BengaliSupplement
gurmukhi	—	Gurmukhi
gujarati	—	Gujarati
oriya	—	Oriya
telugu	—	Telugu
kannada	—	Kannada
malayalam	—	Malayalam
sinhala	—	Sinhala
lao	—	Lao
myanmar	—	Myanmar, MyanmarExtendedA, MyanmarExtendedB
ethiopic	—	Ethiopic, EthiopicExtended, EthiopicExtendedA, EthiopicExtendedB
khmer	—	Khmer, KhmerSymbols
ogham	—	Ogham
cherokee	—	Cherokee, CherokeeSupplement
nko	—	NKo
tifinagh	—	Tifinagh
yi	—	YiSyllables, YiRadicals
adlam	—	Adlam
thaana	—	Thaana
mandaic	—	Mandaic

A Writing Profile identifies Unicode coverage rather than a language-specific font choice. For instance, `chinese`, `japanese` and `korean` overlap in their Han coverage. The older `sanskrit`, `hindi`, `urdu` and `manchu` names remain useful in the registered preset namespace. The `kharoshthi` Profile and resource alias use that spelling; the historical `kharosthi` preset retains its established spelling.

A.8 Font Resource and Pack aliases (85)

The bracketed resource name in a writing configuration is one of the **85 registered resource aliases** below. The corresponding source-family column identifies the legacy catalog family whose font faces and file-resolution rules supply the resource. Several aliases intentionally share a source family but select different faces. An alias can designate a multi-face pack, a script-specific resource, or a particular face.

```
\UseWriting{[libertinus]latin}
\UseWriting{[chinese-traditional]chinese}
\UseWriting{[naskh]arabic}
\Writing{[mnglwhite,writing=vertical]mongolian}{    }
```

Resource aliases and legacy Preset IDs occupy separate namespaces. In particular, `basic` selects the Noto Western resource associated with the legacy `noto` family, whereas `\UseFont{noto}` continues to use the legacy preset. `\UseFont{arabic}` follows its complete historical font recipe; `\UseWriting{[naskh]arabic}` independently selects the `naskh` resource for the `arabic` Unicode Profile.

Resource/pack alias	Source catalog family
basic	noto
cmu	cmu
times	times
gentium	gentium
charis	charis
libertinus	libertinus
latinmodern	mlmodern
chinese-simplified	chinese_simplified
chinese-traditional	chinese_traditional
japanese	japanese
korean	korean
wenjin	wenjin
shanggu	shanggu
armenian	armenian
devanagari	devanagari
tamil	tamil
georgian	georgian
tibetan	tibetan
hebrew	hebrew
arabic	arabic
syriac	syriac
syriac-eastern	syriac_eastern
syriac-western	syriac
thai	thai

Resource/pack alias	Source catalog family
thai-looped	thai
naskh	arabic
kufi	arabic
nastaliq	urdu
aref	arabic
davidlibre	hebrew
khawa	tibetan
carian	anatolian
coptic	coptic
cuneiform	cuneiform
glagolitic	glagolitic
old-italic	italic
old-hungarian	hungarian
runic	runic
brahmi	brahmi
kharoshthi	kharosthi
aramaic	aramaic
nabataean	nabataean
phoenician	phoenician
avestan	avestan
pahlavi-parthian	pahlavi_parthian
pahlavi-inscriptional	pahlavi_inscriptional
pahlavi-psalter	pahlavi_psalter
manichaean	manichaean
samaritan	samaritan
sogdian	sogdian
old-sogdian	sogdian_old
old-turkic	turkic
old-uyghur	uyghur
egyptian	egyptian
khitan-small	khitan_small
tangut	tangut
mongolian	mongolian
mnglwhite	mongolian
mngltitle	mongolian
mnglwriting	mongolian
mnglart	mongolian
monbaiti	mongolian_baiti
abxy	manchu
segoeuihistoric	segoe
tiro	sanskrit

Resource/pack alias	Source catalog family
iming	bopomofo
zhuque	shanggu
simsun	sim
simhei	sim
simfang	sim
simkai	sim
arial	times
consolas	times
nomnatong	vietnamese_hannom
minhnguyen	vietnamese_hannom
gothicnguyen	vietnamese_hannom
syrcom	syriac
syrcom-edessa	syriac
syrcom-midyat	syriac
syrcom-jerusalem	syriac
syrcom-jerusalem-bold	syriac
syrcom-adiabene	syriac_eastern
syrcom-ctesiphon	syriac_eastern
tangut-n4694	tangut
new-tangut-std	tangut

The alias list is the set of names recognized by IMPE's writing registry, rather than an inventory of font binaries included in a release archive. Fonts already available in the TEX or system font environment can resolve normally. For other resources, supply the corresponding font files or choose an explicit `font=` or `file=` override. Multiple aliases may resolve to faces in one source family.

A.9 Style-face shorthand aliases (6)

The face-override options `bold=`, `italic=`, `bolditalic=`, `sans=` and the other documented style slots can use the following explicitly registered shorthand face names. These six names are also included among the 85 resource aliases above; this separate table shows the exact selected face file.

Style alias	Font face file
zhuque	ZhuqueFangsong-Regular.ttf
simkai	simkai.ttf
simfang	simfang.ttf
aref	ArefRuqaa-Regular.ttf
naskh	NotoNaskhArabic-Regular.ttf
nastaliq	NotoNastaliqUrdu-Regular.ttf

Style alias	Font face file

For example, `\UseWriting{[shanggu,italic=zhuque]chinese}` maps the italic style explicitly. A style alias is a named face selection, distinct from a separate Writing Profile. Stylemap and declared faces determine the final eight-style family mapping; named overrides take precedence.

A.10 Additional script rendering commands

Besides the primary local commands in the registered-preset table, four legacy vertical forms are generated from families configured for local vertical rotation:

Command	Typesetting behavior
`\MOv[offset]{text}`	Mongolian, local vertical rotation.
`\MObv[offset]{text}`	Mongolian Baiti, local vertical rotation.
`\MCv[offset]{text}`	Manchu, local vertical rotation.
`\UYv[offset]{text}`	Old Uyghur, local vertical rotation.

The optional vertical offset defaults to `0pt`. These commands become available when their corresponding family is loaded. The Egyptian font preset supplies `\EG` and `\IMPEEgyptianText`. For Khitan, `\KHS` is the normal stacked entry point; `\KHSstack` and `\KHSstackblock` offer explicit inline and paragraph-level stack forms. Their XSR rendering requirements are described in the Fonts section.

A.11 Compatibility class, layout and feature names

The following names are retained to support existing source documents. New documents use the canonical classes and features listed first in this appendix.

Compatibility class	Canonical replacement
`impeart_zh`	`impeart` with `lang=zh`
`impereport_zh`	`impereport` with `lang=zh`
`impebook_zh`	`impebook` with `lang=zh`
`impebeamer_zh`	`impebeamer` with `lang=zh`
`nextart`, `nextreport`, `nextbook`, `nextbeamer`	Matching canonical `impe*` wrapper
`next*_zh`	Matching canonical wrapper with `lang=zh`
`nextsystem.sty`	`impe.sty`

Compatibility ID	Current role
`report`	Historical Chinese-report layout alias.
`legacy_*`	Historical 1.0.5 layout parameter presets.
`\UseLegacyLayout`	Select the 1.0.5 layout for the current canonical class/language.
`bib`	Feature alias for `citations`.
`header`	Feature alias for `headers`.
`lists_envs`	Feature for the retained `ExampleBlock` environment.

A.12 Compatibility-only feature helpers

These helpers remain available for existing documents. The native package interfaces listed above are recommended for new authoring:

Legacy interface	Supported purpose
`\Term[options]{display}` `[description]`	Bold first-occurrence term and its historical index entry.
`\IndexTitle`	Original default index-title override.
`TableInlineFit`, `TableLong`	Old fixed-format table environments.
`TableBook*`, `NiceBooktable*`	Old book-style and annotated table helpers.
`OneImage*`, `PanelFigure*`, `\Panel`	Historical figure and multi-panel helpers.
`ExampleBlock`	Historical indented example display.

The old table/figure control commands, column types `L`, `C`, `R`, `P`, `M`, `B` and original image defaults remain available as documented in `docs/FEATURES.md`. Prefer the native package commands above for new documents.

A.13 Advanced extension interfaces

Extension authors can use the catalog-level `\FontRegisterFamily{...}` interface, together with `\LayoutPresetRegister{...}` and `\LayoutPresetDeclare{...}`. Ordinary users should prefer the public setup and native package APIs rather than the underlying internal `\Next...` commands. See `docs/FONTS.md`, `docs/LAYOUTS.md`, `docs/SYSTEM.md` and `docs/EXTENDING.md`.

B Showcase

The following pages reproduce the complete canonical IMPE showcase. The showcase is a visual companion to the reference manual, demonstrating multilingual font routing, complex shaping, right-to-left scripts, vertical writing, CJK regional forms, and combinations of document features.

The source repository provides the current Showcase. The v1.1.0 release asset provides the corresponding versioned PDF.

IMPE LaTeX System

Canonical Typesetting Showcase

YANG Sikai

v1.1.0

Abstract

This document is the canonical visual specimen for the current IMPE LaTeX System public interface. It combines a local-family style audit, a deliberately small Unicode-range routing regression, dense multilingual paragraph composition with explicit local overrides, right-to-left text, vertical writing, generic complex shaping, CJK and Han family selection, and the principal document features provided by IMPE. It is intended both as a public showcase and as a release-time visual regression specimen.[1]

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[1] The showcase is intentionally broader than the focused regression fixtures under `tests/`; those fixtures isolate individual failures.

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1 Local Font Families and Styles

Table 1 pairs concise script specimens with eight corresponding text styles. Repeated specimens make differences between typefaces directly comparable; where a face is not supplied, the registered family's normal fallback applies. Egyptian format controls and Khitan clusters use their native two-dimensional layouts; Inner Asian samples use vertical writing. The matrix continues across landscape pages as needed.

2 Multilingual and Mixed-Script Composition

Written traditions carry voices across languages, centuries, and regions. The passages here bring together reflections on books and learning alongside short proverbial and scriptural lines, composed in their own writing systems.

2.1 Polyglot Composition

The same page can accommodate many scripts without assigning any single one the role of a translation or a transcription. In the following continuous paragraph, five writing systems are set directly in Unicode, while the others retain their chosen local typefaces.

Գրքերը մեզ պատմում են անցյալի մարդկանց մասին: ძველ წიგნებში მრავალი თაობის აზრი და გამოცდილებაა შემონახული. Ογοϩ ⲡⲓⲗⲟⲅⲟⲥ ⲁⲛⲱⲱⲙ, ⲛⲟⲩⲥⲁⲣⲝ ⲟⲩⲟϩ ⲁⲛⲱⲱⲙ ⲛⲏⲏⲧⲉⲛ. 𓅱𓏏 𓊪𓎛𓐍𓄿𓂋 𓇋𓂋𓏏𓈙 𓐍𓄿𓏏𓏏𓈙, 𓅓 𓐍𓄿𓏏𓏏𓈙 𓇋𓂋𓏏𓈙 𓎛𓏏 𓇋𓈙𓋴𓏏. विद्या ददाति विनयं । கற்க கசடறக் கற்பவை கற்றபின் நிற்க அதற்குத் தக. ཤེས་རབ་དང་སྦྱིང་རྗེའི་མིའི་ཚོས་ཀྱི་རྒྱ་ཡིན། الكتاب خير جليس، والعلم نور. ہماری کتابیں ہمیں ماضی کی دنیا سے ملاتی ہیں۔ החכמה טובה

Table 1: Typeface specimens in eight styles.

Typeface	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Latin typefaces								
CMU Serif	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	Latina	**Latina**
Noto	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
Times	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
Gentium Plus	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
Charis SIL	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
Libertinus	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	**Latina**
MLModern	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
European and Caucasian scripts								
Carian	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ
Coptic	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ
Glagolitic	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ
Old Italic	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I
Old Hungarian	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ
Runic	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬
Armenian	հայերեն	**հայերեն**	հայերեն	**հայերեն**	հայերեն	**հայերեն**	հայերեն	**հայերեն**
Georgian	ქართული	**ქართული**	ქართული	**ქართული**	ქართული	**ქართული**	ქართული	**ქართული**
South Asian and Tibetan scripts								
Devanagari ·Hindi	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**
Devanagari ·Sanskrit	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी
Devanagari	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**
Tamil	தமிழ்	**தமிழ்**	*தமிழ்*	***தமிழ்***	தமிழ்	**தமிழ்**	தமிழ்	**தமிழ்**

Typeface	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Brahmi	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰
Brahmi ·Segoe	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰
Kharoshthi	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿
Tibetan	བོད་ཡིག	**བོད་ཡིག**	ཨོྃ་ཨཱཿ	**ཨོྃ་ཨཱཿ**	བོད་ཡིག	**བོད་ཡིག**	ཨོྃ་ཨཱཿ	**ཨོྃ་ཨཱཿ**
Thai	ภาษาไทย	**ภาษาไทย**	ภาษาไทย	**ภาษาไทย**	ภาษาไทย	**ภาษาไทย**	ภาษาไทย	**ภาษาไทย**
West Asian and Iranian scripts								
Arabic	العربية	العربية	العربية	العربية	العربية	**العربية**	العربية	**العربية**
Arabic ·Nastaliq	نستعليق	نستعليق	نستعليق	نستعليق	نستعليق	نستعليق	نستعليق	نستعليق
Imperial Aramaic	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀
Nabataean	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂
Hebrew	עברית	**עברית**	עברית	**עברית**	עברית	**עברית**	עברית	**עברית**
Syriac	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**
Syriac ·Eastern	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**
Inscriptional Parthian	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠
Inscriptional Pahlavi	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠
Psalter Pahlavi	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂
Avestan	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀
Manichaean	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀
Phoenician	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌
Samaritan	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ
Sogdian	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤
Old Sogdian	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆

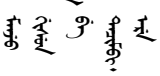
Typeface	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Ancient Near Eastern scripts								
Cuneiform	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿
Egyptian Hieroglyphs	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝
East Asian scripts and Han families								
Bopomofo	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ
Han ·Simplified	汉字	**汉字**	汉字	**汉字**	汉字	**汉字**	汉字	**汉字**
Han ·Traditional	漢字	**漢字**	漢字	**漢字**	漢字	**漢字**	漢字	**漢字**
Han ·Hiragana · Katakana	漢字かなカナ	**漢字かなカナ**	漢字かなカナ	**漢字かなカナ**	漢字かなカナ	**漢字かなカナ**	漢字かなカナ	**漢字かなカナ**
Han ·WenJin	漢字	漢字	漢字	漢字	漢字	漢字	漢字	漢字
Han ·Shanggu	漢字	**漢字**	漢字	**漢字**	**漢字**	**漢字**	**漢字**	**漢字**
Han ·Sim	汉字	汉字	汉字	汉字	汉字	汉字	汉字	汉字
Hangul ·Han	한글漢字	**한글漢字**	한글漢字	**한글漢字**	한글漢字	**한글漢字**	한글漢字	**한글漢字**
Tangut	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦
Inner Asian scripts								
Khitan Small Script	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列
Mongolian	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ
Mongolian Baiti	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ
Manchu	ᠮᡠᠨ	ᠮᡠᠨ	ᠮᡠᠨ	ᠮᡠᠨ	ᠮᡠᠨᡤᡠᠯ	ᠮᡠᠨᡤᡠᠯ	ᠮᡠᠨᡤᡠᠯ	ᠮᡠᠨᡤᡠᠯ
Old Turkic	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇
Old Uyghur	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏

Typeface	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Vietnamese writing traditions								
Latin ·Quốc Ngữ	Quốc ngữ	**Quốc ngữ**	Quốc ngữ	**Quốc ngữ**	Quốc ngữ	Quốc ngữ	Quốc ngữ	Quốc ngữ
Han ·Chữ Nôm	笀喃	**笀喃**	笀喃	笀喃	**笀喃**	**笀喃**	**笀喃**	笀喃

.מכסף ומזהב .𐡀𐡋𐡋𐡄 𐡀𐡅𐡌 ,𐡌𐡏𐡃𐡓𐡀 𐡇𐡎𐡔𐡄 .𐡇𐡊𐡄 𐡆𐡍 𐡌𐡏𐡀𐡍𐡉 𐡀𐡎𐡕𐡄 𐡄𐡊 ᠣᠯᠠᠨ ᠤᠨᠰᠢᠭᠰᠠᠨ ᠬᠡᠪᠴᠢᠶᠡᠨ ᠤ ᠳᠡᠭᠡᠨ, Dǰdl. 古い図書館でカタログをめくり、昔の人の手紙を読みました。 古典을 읽으면 오늘의 질문을 다시 생각하게 됩니다. 舊書中的一句話，有時會在數百年後重新被人理解。

2.2 Mixed Writing Directions

A page may also alternate between horizontal sentences and vertically written words. Here the Mongolian and Manchu passages rise between ordinary horizontal lines, while each writing system retains its own orientation.

翻開一本舊書，紙頁上的文字把我們帶到另一個時代。  古い手紙を読むと、文字の形から書き手の暮らしまで想像できます。  오래된 기록을 읽는 일은 낯선 사람의 목소리에 귀를 기울이는 일입니다. 文字的方向雖然不同，閱讀仍讓相隔久遠的人彼此相遇。

3 Right-to-Left Scripts

Arabic, Urdu, Hebrew, and Syriac begin their lines at the right-hand edge, yet their books also accommodate dates, shelfmarks, and occasional passages in other writing systems. These pages bring together modern prose, verse, and older scriptural traditions.

3.1 Arabic

In a quiet manuscript room, a catalogue entry may be the beginning of a much longer story.

في قاعة المخطوطات رفوف من الخشب الداكن، وعلى الرفوف كتب نُسخت في مدن مختلفة وعصور متباعدة. تحمل بعض الصفحات أسماء ناسخين وقرّاء، وتحفظ الهوامش أسئلة صغيرة تركها أشخاص لا نعرف عن حياتهم إلا القليل. تُظهر إحدى النسخ آثار التصحيحات التي أضافها ناسخ لاحق، وتروي حواشيها كيف انتقل الكتاب بين أيدي القرّاء. وحين يقارن الباحث نسختين من الكتاب نفسه يجد أن اختلاف كلمة واحدة قد يكشف طريق النص من مدينة إلى أخرى. لهذا لا تكون القراءة عودة إلى الماضي وحده؛ إنها حوار جديد مع أناس حفظوا الكلمات ونقلوها إلى من جاء بعدهم.

A fully vocalized couplet by al-Mutanabbī offers another typographic rhythm, with its vowel marks and joined letterforms:

أَعَزُّ مَكَانٍ فِي الدُّنَى سَرْجُ سَابِحٍ، وَخَيْرُ جَلِيسٍ فِي الزَّمَانِ كِتَابُ.

3.2 Urdu

ایک پرانے کتب خانے میں داخل ہوتے ہی کاغذ اور روشنائی کی ایک مانوس خوشبو محسوس ہوتی ہے۔ لکڑی کی الماریوں میں رکھی ہوئی کتابوں پر زمانے کے نشان صاف دکھائی دیتے ہیں، لیکن ان کے صفحات آج بھی نئے سوال پیدا کرتے ہیں۔ بعض نسخوں کے حاشیوں میں پچھلے قارئین نے اپنے خیالات لکھے ہیں؛ کہیں کسی مشکل لفظ کا مطلب درج ہے اور کہیں صرف ایک چھوٹا سا سوالیہ نشان۔ سن ۱۹۲۷ کی ایک یادداشت میں کتاب کے مالک نے لکھا تھا کہ اسے یہ نسخہ سفر کے دوران ملا۔ اب وہی کتاب دوسرے شہر کی لائبریری میں موجود ہے، جہاں ہر نیا قاری اس کی کہانی میں اپنا حصہ شامل کرتا ہے۔ یوں ایک کتاب صرف الفاظ کا مجموعہ نہیں رہتی بلکہ مختلف زمانوں اور لوگوں کے درمیان

رابطے کا ذریعہ بن جاتی ہے۔

3.3 Hebrew

כשנכנסים לספרייה ישנה, אפשר לפעמים לשמוע את רשרוש הדפים יותר מכל קול אחר. על שולחן העץ מונחים כתבי יד, מהדורות מודפסות ומחברת של קורא שכבר איננו. בשוליו של אחד הספרים נכתבה הערה קצרה בשנת 1896, ולידה סימן שאלה קטן. כעבור דורות פותח קורא אחר את אותו העמוד וממשיך את השיחה שהתחילה לפניו. כך נשמרות בספרייה לא רק המילים שבספרים, אלא גם עקבותיהם של האנשים שקראו אותם, העירו עליהם והעבירו אותם לידיים אחרות.

The opening of Genesis presents a different Hebrew page: fully pointed consonants, cantillation marks, and the verse-ending punctuation.

בְּרֵאשִׁית בָּרָא אֱלֹהִים אֵת הַשָּׁמַיִם וְאֵת הָאָרֶץ׃

3.4 Syriac

The opening of the Gospel of John in the Syriac Peshitta places words and light at the heart of its narrative. The following continuous passage is taken from John 1:2–5.

ܗܘ ܗܘܐ ܒܪܫܝܬ ܥܡ ܐܠܗܐ ܟܠ ܫܪܝܪ ܗܘ ܡܠܬܐ ❖ ܘܡܠܬܐ ܗܘ ܗܘܬ ܠܘܬ ܐܠܗܐ ❖ ܟܠ ܡܕܝܢܬܐ ܒܗ ܐܬܒܪܝ ❖ ܘܒܚܫܘܟܐ ❖ ܢܗܝܪܐ ܒܚܫܘܟܐ ❖ ܘܚܫܘܟܐ ܠܐ ܐܕܟܝܗ

Two traditional typefaces set the same opening verse, allowing their letterforms to be read alongside one another.

General Syriac

ܗܘ ܗܘܐ ܒܪܫܝܬ ܥܡ ܐܠܗܐ ❖ ܟܠ ܡܕܝܢܬܐ ܒܗ ܐܬܒܪܝ ❖ ܘܒܚܫܘܟܐ ❖ ܢܗܝܪܐ ܒܚܫܘܟܐ ❖ ܘܚܫܘܟܐ ❖ ܠܐ ܐܕܟܝܗ

Eastern Syriac

ܗܘ ܗܘܐ ܒܪܫܝܬ ܥܡ ܐܠܗܐ ❖ ܟܠ ܡܕܝܢܬܐ ܒܗ ܐܬܒܪܝ ❖ ܘܒܚܫܘܟܐ ❖ ܢܗܝܪܐ ❖ ܘܚܫܘܟܐ ❖ ܠܐ ܐܕܟܝܗ

4 Vertical Writing

Mongolian, Manchu, and Old Uyghur are written in vertical lines. In a horizontal IMPE document, local vertical runs retain that orientation while individual word-sized units can wrap within the surrounding page. The following passages compare the two settings and their different letterforms.

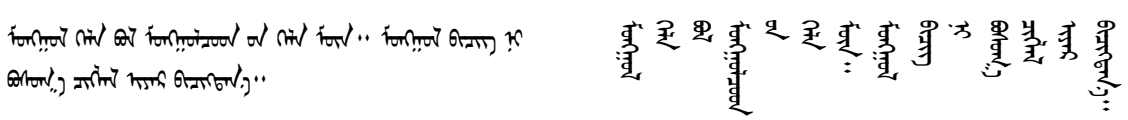
4.1 Mongolian

The same Mongolian passage appears in horizontal and vertical settings.

Horizontal reference

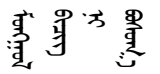
᠂ᠪᠣᠯᠣᠭᠠᠨ ᠤᠨ ᠲᠡᠮᠳᠡᠭ ᠪᠣᠯᠣᠭᠠᠨ ᠤᠨ ᠤᠨ ᠪᠣᠯᠣᠭᠠᠨ ᠂᠂ ᠪᠣᠯᠣᠭᠠᠨ ᠦᠰᠦᠭ ᠢ
ᠲᠡᠮᠳᠡᠭᠯᠡᠵᠦ ᠴᠢᠳᠠᠵᠤ ᠪᠣᠯᠣᠭᠠᠨ᠂᠂

Vertical

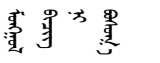


The regular Mongolian family and Mongolian Baiti give the same short phrase different appearances in vertical setting:

Mongolian



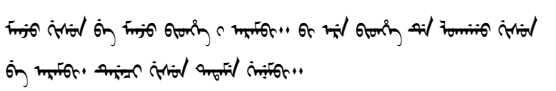
Mongolian Baiti



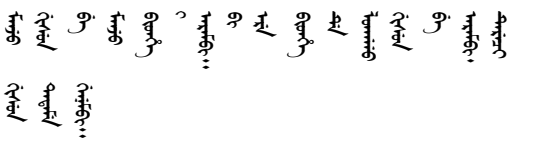
4.2 Manchu

The same Manchu passage appears in both orientations, allowing direct comparison without changing the text.

Horizontal reference



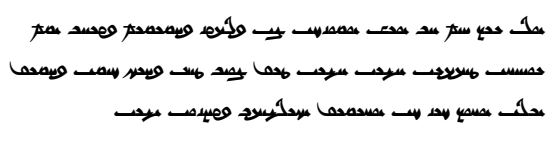
Vertical



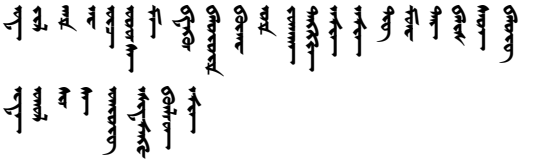
4.3 Old Uyghur

A provisional Unicode rendering based on the published transliteration of the colophon of the Turfan manuscript U 338 supplies a continuous Old Uyghur passage. It is presented as a typographic specimen rather than a critical edition.

Horizontal reference



Vertical



5 Complex Shaping and Stacked Sequences

Two kinds of composition meet on these pages. Devanagari, Tibetan, Brahmi, Kharoshthi, and Thai rely on the shaping of ordinary Unicode text by the selected OpenType fonts. Egyptian Hieroglyphs and Khitan Small Script instead use XSR to arrange signs into two-dimensional units while preserving their original Unicode sequences.

5.1 Devanagari

Devanagari makes conjunct consonants, vowel signs, and the virama visible in the shapes of its syllables. The Sanskrit family sets the forms in Table 2 with the typeface used in the Sanskrit verse.

The first three verses of the *Bhagavadgītā* (1.1–3) give a longer, continuous Sanskrit specimen with conjuncts and vowels.

> धृतराष्ट्र उवाच । धर्मक्षेत्रे कुरुक्षेत्रे समवेता युयुत्सवः । मामकाः पाण्डवाश्चैव किमकुर्वत सञ्जय ॥१ ॥ सञ्जय उवाच । दृष्ट्वा तु पाण्डवानीकं व्यूढं दुर्योधनस्तदा । आचार्यमुपसङ्गम्य राजा वचनमब्रवीत् ॥२ ॥ पश्यैतां पाण्डुपुत्राणामाचार्य महतीं चमूम् । व्यूढां द्रुपदपुत्रेण तव शिष्येण धीमता ॥३ ॥

5.2 Tibetan

Tibetan syllables can contain stacked consonants within a single *tsheg*-delimited unit. Table 3 compares short forms; the traditional Four Immeasurables prayer provides a longer passage.

སེམས་ཅན་ཐམས་ཅད་བདེ་བ་དང་བདེ་བའི་རྒྱུ་དང་ལྡན་པར་གྱུར་ཅིག ། སེམས་ཅན་ཐམས་ཅད་སྡུག་བསྔལ་དང་སྡུག་བསྔལ་གྱི་རྒྱུ་དང་བྲལ་བར་གྱུར་ཅིག ། སེམས་ཅན་ཐམས་ཅད་སྡུག་བསྔལ་མེད་པའི་བདེ་བ་དང་མི་འབྲལ་བར་གྱུར་ཅིག ། སེམས་ཅན་ཐམས་ཅད་ཉེ་རིང་ཚགས་སྡང་དང་བྲལ་བའི་བཀྲང་སྙོམ་ལ་གནས་པར་གྱུར་ཅིག།

Table 2: Devanagari shaping specimens

Sequence	Specimen
kṣa	क्ष
jña	ज्ञ
śra	श्र
stra	स्त्र
ddhya	द्ध्य
dharma	धर्म
dharmakṣetre	धर्मक्षेत्रे
saṃskṛtam	संस्कृतम्
oṃ	ॐ

Table 3: Tibetan consonant-stack specimens

Transliteration	Specimen
bsgrubs	བསྒྲུབས་
brgyud	བརྒྱུད་
bstan	བསྟན་
rgyal	རྒྱལ་
dbyangs	དབྱངས་

Sanskrit transcribed in Tibetan letters. The *Heart Sūtra* mantra illustrates Sanskrit consonant stacks, long-vowel marks, anusvāra, and the *svāhā* cluster without switching away from the Tibetan-script font:

ཏྒྱཱ་ག་ཏེ་ག་ཏེ་པཱ་ར་ག་ཏེ། པཱ་ར་སཾ་ག་ཏེ། བོ་དྷི་སྭཱ་ཧཱ།

5.3 Brahmi, Kharoshthi, and Thai

The complete Rummindei (Lumbini) pillar edict of Aśoka, in Prakrit, provides a genuine Brāhmī inscription. The same Unicode transcription is set in two regular fonts; the original inscription's line breaks are joined into one flowing paragraph without changing its characters.

Brāhmī — Noto Sans Brahmi

𑀦𑀺𑀬𑀢𑁆𑀢 𑀓𑀮𑀸𑀓𑁂 𑀫𑀳𑀸𑀬𑀸𑀦𑀲𑁆𑀫𑀺𑀁 𑀯𑀘𑁆𑀘𑀺𑀦𑀻 𑀦𑀸𑀫 𑀭𑀸𑀚𑀓𑀼𑀫𑀸𑀭 𑀓𑀸𑀮𑀸𑀪𑀺𑀲𑀺𑀢𑁂𑀦 𑀯𑁂𑀤𑀸𑀪𑀺𑀕𑀫𑀦𑀸𑀦𑀼𑀪𑀯𑁂𑀦 𑀲𑀫𑁆𑀫𑀸𑀲𑀫𑁆𑀩𑀼𑀥𑁄 𑀮𑀼𑀁𑀩𑀺𑀦𑀻 𑀕𑀸𑀫𑀺𑀦𑀻 𑀉𑀩𑀮𑀺𑀓𑀸 𑀓𑀭𑀸𑀪𑀸𑀓𑀸 𑀲𑀺𑀮𑀸 𑀣𑀽𑀪𑁄 𑀓𑀢𑁄

Brāhmī — Segoe UI Historic

𑀦𑀺𑀬𑀢𑁆𑀢 𑀓𑀮𑀸𑀓𑁂 𑀫𑀳𑀸𑀬𑀸𑀦𑀲𑁆𑀫𑀺𑀁 𑀯𑀘𑁆𑀘𑀺𑀦𑀻 𑀦𑀸𑀫 𑀭𑀸𑀚𑀓𑀼𑀫𑀸𑀭 𑀓𑀸𑀮𑀸𑀪𑀺𑀲𑀺𑀢 𑀫𑁂𑀤𑀸𑀪𑀺𑀕𑀫𑀦𑀸𑀦𑀼𑀪𑀯𑁂𑀦 𑀲𑀫𑁆𑀫𑀸𑀲𑀫𑁆𑀩𑀼𑀥𑁄 𑀮𑀼𑀁𑀩𑀺𑀦𑀻 𑀕𑀸𑀫𑀺𑀦𑀻 𑀉𑀩𑀮𑀺𑀓𑀸 𑀓𑀭𑀸𑀪𑀸𑀓𑀸 𑀲𑀺𑀮𑀸 𑀣𑀽𑀪𑁄 𑀓𑀢𑁄

Source: Aśoka's Rummindei pillar edict, based on E. Hultzsch; Unicode transcription at Brahmi script: Birthplace of the historical Buddha. Segoe UI Historic is a separately installed Microsoft font.

Kharoshthi — illustrative script transcription

𐨑𐨪𐨁𐨐𐨪𐨁 𐨑𐨁𐨐𐨪𐨁 𐨑𐨁𐨐𐨪𐨁

This short Kharoshthi passage is a modern script conversion of Sanskrit, not an attested ancient Gāndhārī inscription. Its right-to-left shaping is the focus of this specimen.

Thai — continuous contemporary prose

เมื่อเปิดหนังสือเก่าในห้องสมุด เราอาจพบลายมือของผู้อ่านจากหลายยุคสมัย บางคนขีดเส้นใต้ข้อความที่ชอบ บางคนเขียนคำถามไว้ริมหน้า และบางคนบันทึกวันเวลาที่อ่านจบ เรื่องราวเหล่านี้ทำให้หนังสือเล่มเดียวเป็นเหมือนบทสนทนาที่ดำเนินต่อไปข้ามกาลเวลา

5.4 Egyptian Hieroglyphic Composition

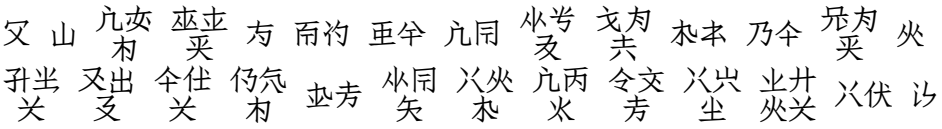
The opening of the *Tale of the Shipwrecked Sailor*, lines 1–15 of Bob Richmond's horizontal UMdC transcription, is set as one continuous flowing Egyptian passage. The manuscript itself is in hieratic; the presentation below uses hieroglyphic transcription. Editorial line numbers and colour markup have been removed, while the original Unicode hieroglyphic signs and format controls remain.



Source: UMdC, ShipwreckedSailorHorizontal.umdc, opening lines 1–15; adapted from JSesh examples with reference to James P. Allen.

5.5 Khitan Small Script

The opening of the 1134 *Record of the Langjun*, an attested Khitan Small Script inscription, is transcribed in two-dimensional Unicode clusters. The first two lines of the Wikisource transcription form one flowing paragraph here; original transcription line breaks are omitted and cluster boundaries are preserved.



Source: Wikisource, 大金皇弟都統經略郎君行記, opening two transcription lines (Unicode transcription, CC BY-SA). This is a horizontal setting of a traditionally vertical inscription.

6 East Asian Writing and CJK Family Selection

This chapter compares document-level CJK typography with explicit local families for Chinese, Korean, Japanese, Vietnamese, and Tangut. The same Han codepoints may take different regional forms.

6.1 Chinese

Four paragraphs distinguish Simplified Chinese, mainland-style Traditional Chinese, Taiwan Traditional Chinese, and the document's Shanggu family. The three Traditional Chinese specimens use exactly the same source text.

Simplified Chinese — Noto SC 秋天的清晨，图书馆刚刚开门。书架上放着一本旧地图，画有山谷、城门与河流，旁边是《龙城纪事》。读书的人翻开书页，读到国王命令学者修订典籍的故事：有人研究龙骨，也有人制作木具，记下直角的形状。窗外下起雨，墨色在灯下格外清晰。

Traditional Chinese — Noto SC 秋天的清晨，圖書館剛剛開門。書架上放著一本舊地圖，畫有山谷、城門與河流，旁邊是《龍城紀事》。讀書的人翻開書頁，讀到國王命令學者修訂典籍的故事：有人研究龍骨，也有人製作木具，記下直角的形狀。窗外下起雨，墨色在燈下格外清晰。

Traditional Chinese — Noto TC 秋天的清晨，圖書館剛剛開門。書架上放著一本舊地圖，畫有山谷、城門與河流，旁邊是《龍城紀事》。讀書的人翻開書頁，讀到國王命令學者修訂典籍的故事：有人研究龍骨，也有人製作木具，記下直角的形狀。窗外下起雨，墨色在燈下格外清晰。

Traditional Chinese — Shanggu (document default) 秋天的淸晨，圖書館剛剛開門。書架上放著一本舊地圖，畫有山谷、城門與河流，旁邊是《龍城紀事》。讀書的人翻開書頁，讀到國王命令學者修訂典籍的故事：有人研究龍骨，也有人製作木具，記下直角的形狀。窗外下起雨，墨色在燈下格外淸晰。

6.2 Korean and Historical Hangul

The first two passages compare Hangul-only prose with a mixed Hanja–Hangul register. Both use the Korean family.

Modern Korean 가을비가 그친 오후, 작은 도서관의 창가에는 햇빛이 조용히 들어왔다. 학생들은 오래된 지도를 펼쳐 놓고 사라진 마을의 이름을 찾아보았다. 한 사람은 책의 여백에 질문을 적었고, 다른 사람은 다음 주에 다시 와서 자료를 읽기로 했다.

Mixed Hanja and Hangul 古書를 정리하던 학생은 朝鮮時代의 日記 한 권을 발견했다. 紙面에는 한글과 漢字가 나란히 적혀 있었고, 여러 사람의 손으로 남긴 註釋도 보였다. 학생은 낡은 책을 조심스럽게 넘기며 당시의 生活과 言語를 상상했다.

Middle Korean — *Hunminjeongeum eonhae* This continuous excerpt from the *Hunminjeongeum eonhae* retains arae-a, archaic finals, and tone marks. Its Unicode input includes precomposed modern syllables followed by archaic final jamo: the local Hangul shaper must compose these just as it composes their canonically equivalent decomposed sequences.

나·랏:말ᄊᆞ미 中듕國·귁·에달·아 文문字·쭝·와·로서르ᄉᆞᄆᆞᆺ·디아·니ᄒᆞᆯ·씨 ·이런젼·ᄎᆞ·로어·린 百·빅姓·셩·이니르·고·져·홇·배이·셔·도

6.3 Japanese

Ordinary modern Japanese combines kanji, hiragana, katakana, punctuation, and regionally selected Han forms within the same prose paragraph.

春の朝、京都の小さな書店に新しい本が届いた。店主は箱を開け、「歴史と文化」の棚に本を並べた。古い地図には城門や学校が描かれ、表紙には龍の絵がある。来店した学生はカタログで目当ての一冊を探し、静かな店内でページをめくった。

6.4 Vietnamese

These parallel excerpts reproduce the opening of the 2013 Vietnamese Constitution's preamble. The same Quốc Ngữ text appears first in the document's Libertinus family and then in the dedicated Vietnamese family. The Hán-Nôm specimen is a modern transcription of the same passage, not a historical constitutional manuscript; the local regular font is Minh Nguyen.

Quốc Ngữ — Libertinus Trải qua mấy nghìn năm lịch sử, Nhân dân Việt Nam lao động cần cù, sáng tạo, đấu tranh anh dũng để dựng nước và giữ nước, đã hun đúc nên truyền thống yêu nước, đoàn kết, nhân nghĩa, kiên cường, bất khuất và xây dựng nên nền văn hiến Việt Nam. Từ năm 1930, dưới sự lãnh đạo của Đảng Cộng sản Việt Nam do Chủ tịch Hồ Chí Minh sáng lập và rèn luyện, Nhân dân ta tiến hành cuộc đấu tranh lâu dài, đầy gian khổ, hy sinh vì độc lập, tự do của dân tộc, vì hạnh phúc của Nhân dân. Cách mạng tháng Tám thành công, ngày 2 tháng 9 năm 1945, Chủ tịch Hồ Chí Minh đọc Tuyên ngôn độc lập, khai sinh ra nước Việt Nam dân chủ cộng hòa, nay là Cộng hòa xã hội chủ nghĩa Việt Nam.

Quốc Ngữ — Vietnamese family Trải qua mấy nghìn năm lịch sử, Nhân dân Việt Nam lao động cần cù, sáng tạo, đấu tranh anh dũng để dựng nước và giữ nước, đã

hun đúc nên truyền thống yêu nước, đoàn kết, nhân nghĩa, kiên cường, bất khuất và xây dựng nên nền văn hiến Việt Nam. Từ năm 1930, dưới sự lãnh đạo của Đảng Cộng sản Việt Nam do Chủ tịch Hồ Chí Minh sáng lập và rèn luyện, Nhân dân ta tiến hành cuộc đấu tranh lâu dài, đầy gian khổ, hy sinh vì độc lập, tự do của dân tộc, vì hạnh phúc của Nhân dân. Cách mạng tháng Tám thành công, ngày 2 tháng 9 năm 1945, Chủ tịch Hồ Chí Minh đọc Tuyên ngôn độc lập, khai sinh ra nước Việt Nam dân chủ cộng hòa, nay là Cộng hòa xã hội chủ nghĩa Việt Nam.

Hán-Nôm — Minh Nguyen Regular 踈過尒酥辭歷史、人民越南勞動勤劬、創造、鬥爭英勇抵踎渃吧符渃、缶熏燤铖傳統悽渃、團結、仁義、堅強、不屈吧埬踎铖垭文獻越南。自辭1930、迂事領導貼黨共產越南由主席胡志明創立吧煓鍊、人民些進行局鬥爭數踴、浩艱苦、犧牲爲獨立、自由貼民族、爲幸福貼人民。革命腳噆成功、暟2腳9辭1945、主席胡志明讀宣言獨立、開生器渃越南民主共和、跉羅共和社會主義越南。

6.5 Tangut

The following paragraph restores the original Unicode Tangut corpus specimen in full, including its editorial punctuation. The text is a composite of short attested sequences from different Tangut works on Wikisource, not a single continuous historical document.

𗍫𗾔𘝦𗏁𗐱𗖵，𘝵𘝵𗏁𗍫𘜶𗏒，𗥃𗖰𗂧𗋢𗊬𘟣，𗏆𗍊𗾇𗐆𗥤𗏵𗖵，𗾛𗖵𗖰𗐱𗐱𗥤，𗾓𗖜𗖰𗍫𗖰𗏵𗏁𗏒𗐱，𗏁𗖵𗤶𗖵𗖵𗖰𗅲𗏁𗖵𗏒，𗍮𗖵𗖰𗍊𗖵𗖰𗏁𗏒，𗾓𗖵𗍫𗖵𗖰𗏁𗏒𗐱，𗾓𗖵𗍫𗖵𗖰𗏁𗏒𗐱，𗏁𗖵𗤶𗖵𗖵𗖰𗅲𗏁𗖵𗏒

Table 4: WenJin CJK block coverage (twenty codepoints per block)

Unicode block and range	Specimen
CJK Unified Ideographs U+4E00--U+9FFF	一丁丂七丄丅丆万丈三上下丌不与丐丑丒专
Extension A U+3400--U+4DBF	丕丙丠个丰乂乃乞乩乭乺乾亂亝亞亟亾亹亿
Extension B U+20000--U+2A6DF	亡亏亿亼亾亏仡且丘丘丣丈丌丗丛丯丳引
Extension C U+2A700--U+2B73F	廾节瓦隼奇奡猡臉氐帡冫夷屆乖秖狩邳枣倢垒
Extension D U+2B740--U+2B81F	丸乖亦爾久灰今侖侮倉傳僜奘岡凅渗前勢卑悊
Extension E U+2B820--U+2CEAF	七不卡东耒匝所轩皂兹趴尧秬杯豽豝戏昱番碖
Extension F U+2CEB0--U+2EBEF	寸才几万刁号丐歹考彐号豕旭尧屯审芬玊卅丆
Extension I U+2EBF0--U+2EE5F	丌舟牛焚仉仿伀仅仿佩价佂侥俣倪俞傕復拿龰
Extension G U+30000--U+3134F	丅丰丘冂巨丸立丐专申屮巡吊砑垚垠冤豺鬲甄
Extension H U+31350--U+323AF	元去陕垚陇來卅浮劾邮爹呇郤郪荚骁皋稠碕篡
Extension J U+323B0--U+3347F	五北冂訐垦葙衅銚锴鄢禾贱卻棚轋灺乖卒吵忝

Table 5: Regional forms of shared Han codepoints

Character	Shanggu	Noto SC	Noto TC	Japanese	Korean
骨	骨	骨	骨	骨	骨
直	直	直	直	直	直
令	令	令	令	令	令
具	具	具	具	具	具
角	角	角	角	角	角
門	門	門	門	門	門
國	國	國	國	國	國
學	學	學	學	學	學
龍	龍	龍	龍	龍	龍

lated formulae as a single derivation.

For a positive-definite matrix, a Gaussian integral evaluates to

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \qquad \det A = 4, \quad \beta > 0,$$

$$\begin{aligned} \mathcal{Z}(\beta) &= \int_{\mathbb{R}^3} \exp\!\left(-\frac{\beta}{2}\mathbf{x}^{\mathsf{T}} A \mathbf{x}\right) \mathrm{d}^3\mathbf{x} \\ &= \left(\frac{2\pi}{\beta}\right)^{3/2} \frac{1}{\sqrt{\det A}} = \frac{1}{2}\left(\frac{2\pi}{\beta}\right)^{3/2}. \end{aligned}$$

For $k \in \mathbb{N}_0$, the residue theorem provides a series and a contour integral:

$$f_k(z) = \frac{e^z}{z^{k+1}} = \sum_{n=0}^{\infty} \frac{z^{n-k-1}}{n!},$$

$$\frac{1}{2\pi i}\oint_{|z|=1} f_k(z)\,\mathrm{d}z = \operatorname{Res}_{z=0} f_k(z) = \frac{1}{k!}.$$

For integrable f, its Fourier transform illustrates wide accents:

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi}\,\mathrm{d}x, \qquad \overline{\widehat{f}(\xi)} = \widehat{\bar{f}}(-\xi).$$

A smooth flat function demonstrates a cases environment:

$$g(x) = \begin{cases} e^{-1/x^2}, & x > 0, \\ 0, & x \le 0, \end{cases} \qquad g^{(r)}(0) = 0 \quad (r = 0, 1, 2, \ldots).$$

Chebyshev's inequality and the decomposition of exterior powers supply independent examples of further mathematical notation:

$$\underbrace{\mathbb{P}(|X - \mathbb{E}[X]| \ge t)}_{\text{tail probability}} \le \frac{\operatorname{Var}(X)}{t^2}, \qquad t > 0.$$

$$\bigwedge^{k}(V \oplus W) \cong \bigoplus_{p+q=k} \left(\bigwedge^{p} V\right) \otimes \left(\bigwedge^{q} W\right).$$

A short inline comparison shows how the same summation and fraction occupy different amounts of vertical space:

Default display style $\displaystyle\sum_{j=1}^{n} \frac{1}{j^2}$

Local text style $\sum_{j=1}^{n} \frac{1}{j^2}$

7.2 Drawing

Figure 1 shows a Gaussian function with its formula and graduated axes. Figure 2 places a schematic manuscript-copying tree within the page.

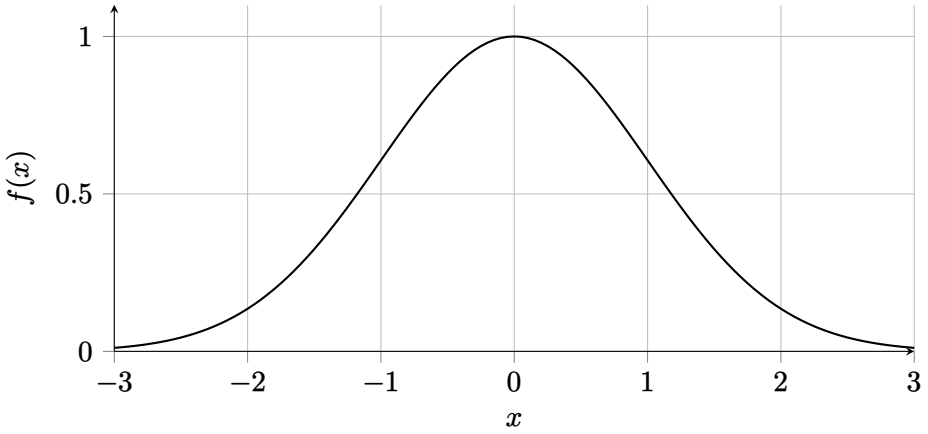


Figure 1: The Gaussian profile $f(x) = e^{-x^2/2}$.

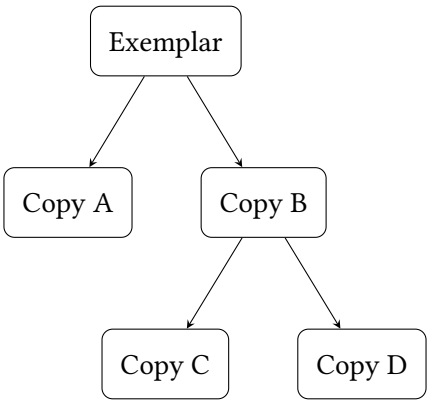


Figure 2: A schematic manuscript-copying tree, not a historical stemma.

7.3 Tables

Table 6 uses the native `table`, `tabularx`, and `booktabs` interfaces loaded by the `tables` feature.

Table 6: Feature integration summary

Feature	Public route	Exercised here
Mathematics	`math`	yes
Tables	`tables`	yes
Images	`image`	yes
Inline glyphs	`glyphs`	yes
Drawing	`drawing`	yes
Citations	`citations`	yes
Index	`index`	yes
Hyperlinks	`hyperlinks`	yes
Running headers	`headers`	yes

7.4 Lists and Quoted Examples

Ordinary lists and quotations do not require an IMPE-specific environment:

1. automatic font ownership;
2. script-specific behavior where required;
3. reusable document features;
4. stable public interfaces.

The native `quote` environment supplies an indented specimen:

> *A canonical showcase should be readable as a document, useful as a visual regression specimen, and representative of the public interface that new documents are expected to use.*

7.5 Images

The `image` feature prepares standard graphics and caption packages. Figure 3 includes a bundled vector image without relying on optional example-image files from the user's TeX distribution. Figure 3 also exercises native cross-references.

7.6 Inline Image Glyphs

A short passage from Chu bamboo manuscripts sets individual manuscript signs directly within the running text. The images function as characters, not floating figures; the transcriptions and source references are reproduced without adopting any particular decipherment.

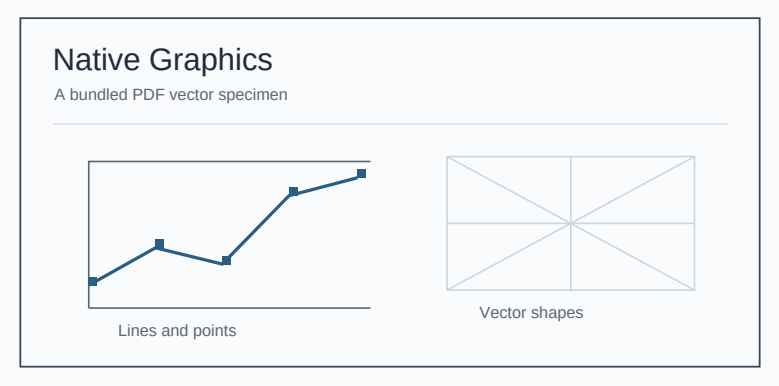


Figure 3: Native graphics inclusion using a bundled vector specimen.

「忮」字的古文字字形爲「  」（《郭店簡・語叢四》，15.14）、「  」（《上博簡・曹沫之陳》，30.16），一般隸定爲「忮」或「攲」。在出土楚地簡帛文獻中，本字的出現如下：

1 凡敓之道，級者爲首。既得其級，言必又及$_{=五}$之而弗亞，必隶其古。隶之而悘，必忮鉿$_{=}$。$_{十五}$之$_{=}$而不可，必𠬜以訛，毋命智我。皮邦芒$_{六}$牏，流澤而行。$_{七}$（《郭店簡・語叢四》）[2]

2 ……□厚飤，思爲前行。三行之逡，句見耑兵，忮$_{三十}$（《上博簡・曹沫之陳》）[3]

「鉿」的古文字字形爲「  」，在出土楚系簡帛文獻中僅見於上述文段 1。

7.7 Citations and Bibliography

Citation support is loaded through the feature system and configured before feature activation. A normal textual citation can therefore be written as Knuth (1984), while a parenthetical citation appears as (Knuth, 1984).

The bibliography is printed near the end of the document.

7.8 Index Terms

Earlier sections use the native `\index` command to register concepts at their actual occurrences. The phrase *canonical showcase* is indexed here; native interfaces also provide a hierarchical example. IMPE supplies unique occurrence-level destinations for its default index, while `imakeidx` and `xindy` retain their usual indexing syntax.

7.9 Hyperlinks, Footnotes, and Running Headers

The table of contents is clickable, numbered headings carry stable targets, and section headings can link back toward their table-of-contents entries. The running header above this page is produced by the `headers` feature.

Footnote markers are also bidirectional.[4]

For an external-link specimen, the project repository can be represented with a normal hyperlink: IMPE LaTeX System on GitHub.

[2] 武漢大學簡帛研究中心、荊門市博物館編著：《楚地出土戰國簡冊合集（一）：郭店楚墓竹書》，文物出版社，2011 年，167–168 頁。

[3] 馬承源主編：《上海博物館藏戰國楚竹書（四）》，上海古籍出版社，2004 年，262 頁。

[4] Clicking the footnote marker jumps to this note; the note marker can link back to the original position.

References

Knuth, D. E. (1984). *The texbook*. Addison-Wesley.